

DM IMCQ 3

Spring 2026

Physiology and Biochemistry

DM IMCQ 3

Physiology 5 Questions

A 26-year-old woman comes to the physician because of a 5-month history of recurrent episodes of abdominal discomfort, diarrhea, and ulcers in the oral cavity. Her aunt had similar symptoms and was diagnosed with malabsorption. Vitals are within normal limits. Physical examination shows no abnormalities. An upper GI endoscopy is performed to evaluate the exocrine pancreatic function. She is administered a gut hormone intravenously and the pancreatic juice is collected through an endoscope. This pancreatic juice is found to be low in volume but rich in digestive enzymes and has normal electrolytes concentration. Which of the following additional physiologic effects is most likely caused by this hormone?

- A. Stimulates pepsinogen secretion from the parietal cells of the stomach
- B. Increases sphincter of Oddi contraction
- C. Slows gastric emptying
- D. Decreases gall bladder contraction
- E. Inhibits acid secretion from the parietal cells of the stomach

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SOM.MK.II.BPM2.3.DM.1.PHYS.1099. List the major excitatory and inhibitory motor neurotransmitters and digestive hormones in the GI tract and their key regulatory functions.

An 8-year-old boy is brought to the physician by his mother because of delayed growth, poor performance in the school, and increased appetite. The mother says that compared to his classmates, he is far behind in speaking and reading skills. He is 3 ft 6 in tall (106 cm) and weighs 40 kg (88-lb); BMI is 35.1 (Normal: 19-25). Physical examination shows short stature, muscle weakness, and obesity. Mental examination shows intellectual disability. He is diagnosed with Prader-Willi syndrome. The physician tells the mother that his appetite is increased because of an elevated serum concentration of a hormone, secreted from the stomach. Which of the following gut hormones is most likely elevated in this patient?

- A. Vasoactive intestinal peptide
- B. Gastric inhibitory peptide
- C. Gastrin
- D. Secretin
- E. Ghrelin
- F. Somatostatin

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A 46-year-old man comes to the physician because of a 5-month history of progressive difficulty opening mouth and chewing food. He says that he is unable to fully open his mouth and grind food because of pain. He has a 10-year history of rheumatoid arthritis. Physical examination shows restricted temporomandibular joint movements bilaterally. In this patient, digestion of carbohydrates in the mouth is impaired because of improper mixing of food with saliva. Which of the following salivary enzymes most likely helps this digestive process?

- A. Amylase
- B. Lipase
- C. Carboxypeptidase
- D. Cholesterol esterase
- E. Elastase
- F. Trypsin

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An 81-year-old man comes to the physician because of a 5-month history of abdominal pain and fatigue. He does not have any major illnesses. His vital signs are within normal limits. Physical examination shows pale tongue and nailbeds. Endoscopy shows a 2-cm mass in the body of the stomach near greater curvature. Examination of a biopsy specimen from this mass shows adenocarcinoma. He undergoes a partial gastrectomy in which the entire body of the stomach is resected, sparing the fundal and antral regions. Which of the following alterations in the composition of the gastric juice is most likely in this patient after surgery?

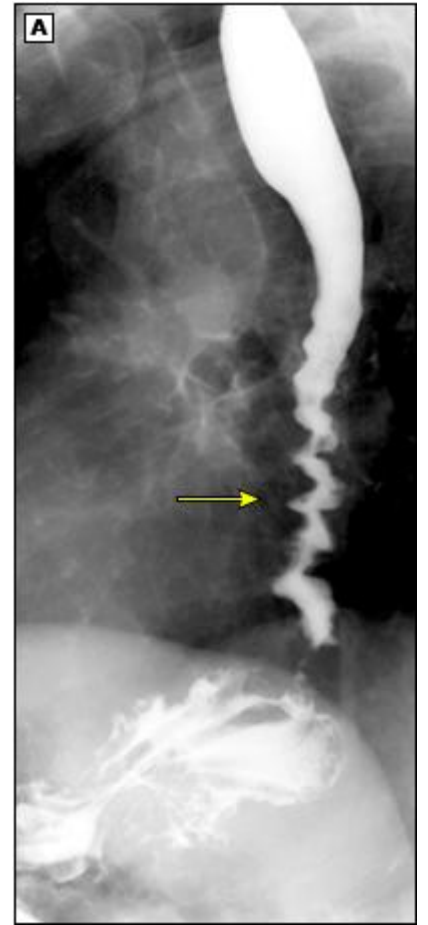
- A. Increased mucin
- B. Decreased pH
- C. Decreased bicarbonate
- D. Reduced intrinsic factor
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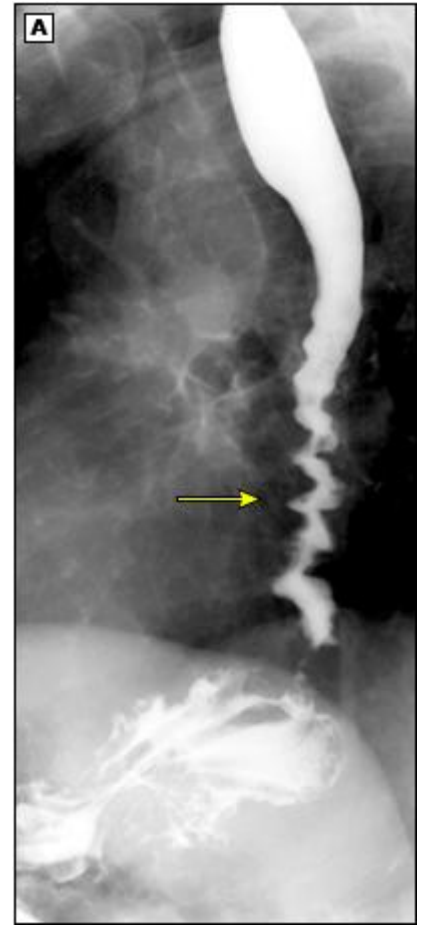
- A. Increased mucin
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A 74-year-old man comes to the physician because of a 6-month history of difficulty swallowing and retrosternal pain. He says that he has a sensation of food getting stuck and passing slowly while swallowing. A barium esophagram shows a “corkscrew” appearance of the lower third of the esophagus as shown by an arrow in the image. Esophageal manometry shows normal lower esophageal sphincter (LES) functions. He is diagnosed with esophageal spasm due to defective functioning of the inhibitory motor neurons of the enteric nervous system in the lower third of the esophagus sparing LES. Which of the following brain centers most likely controls these inhibitory neurons at this site of the esophagus?

- A. Nucleus ambiguus
- B. Dorsal motor nucleus
- C. Area postrema
- D. Nucleus tractus solitarius
- E. Chemoreceptor trigger zone



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Biochemistry 11 Questions

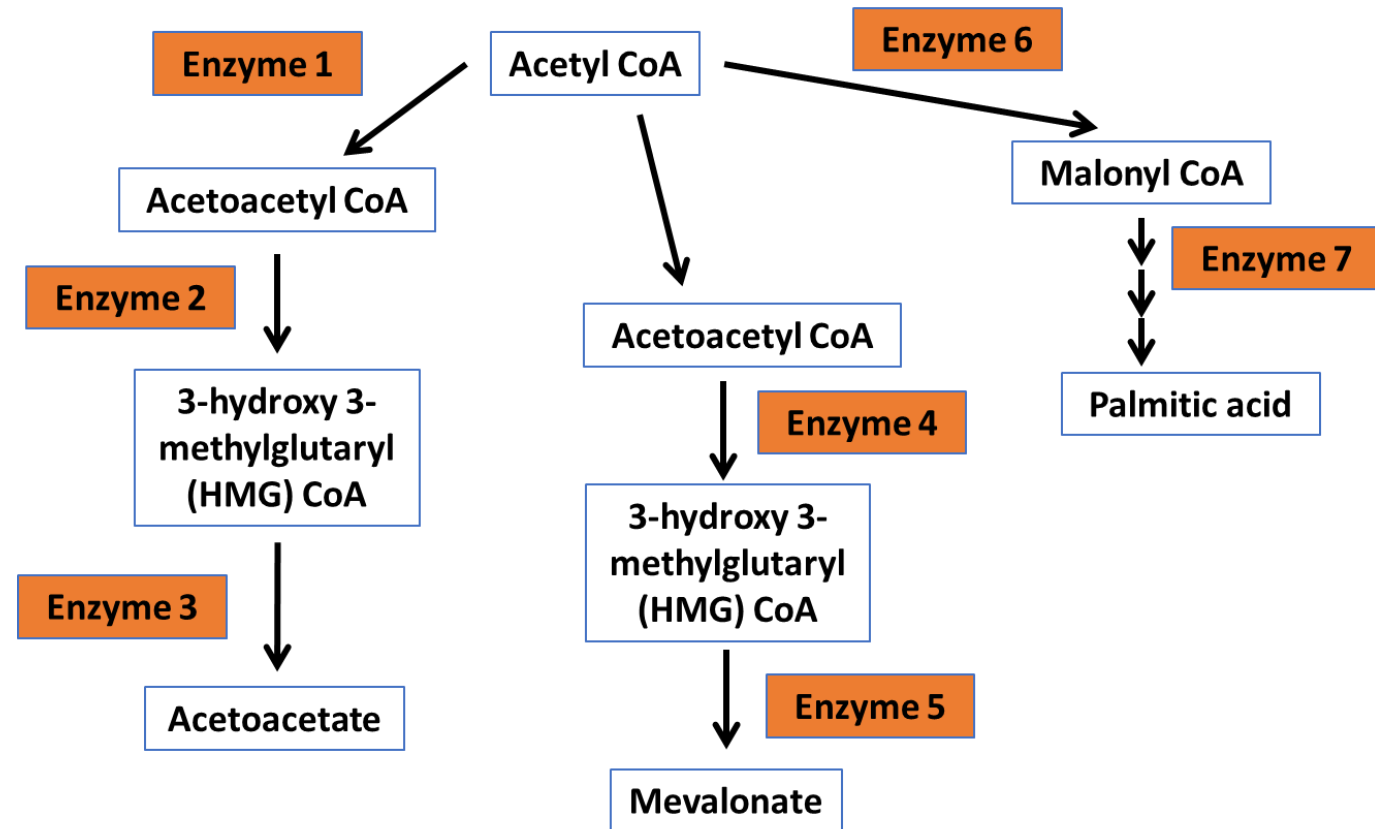
A 25-year-old man comes to the emergency department after participating in a wilderness survival challenge and has been without food for 3 days. He reports mild dizziness and fatigue. Physical examination shows mild dehydration. Laboratory studies show elevated levels of ketone bodies in the blood and urine. Which of the following metabolic changes is most likely in the tissues of this patient?

- A. Increased acetyl CoA carboxylase and hormone sensitive lipase activity
- B. Increased mitochondrial HMG CoA synthase and HMG CoA reductase activity
- C. Increased mitochondrial HMG CoA synthase and decreased AMP-kinase activity
- D. Decreased mitochondrial HMG CoA synthase and increased succinyl CoA-acetoacetate transferase activity
- E. Increased mitochondrial HMG CoA synthase and increased succinyl CoA-acetoacetate CoA transferase activity

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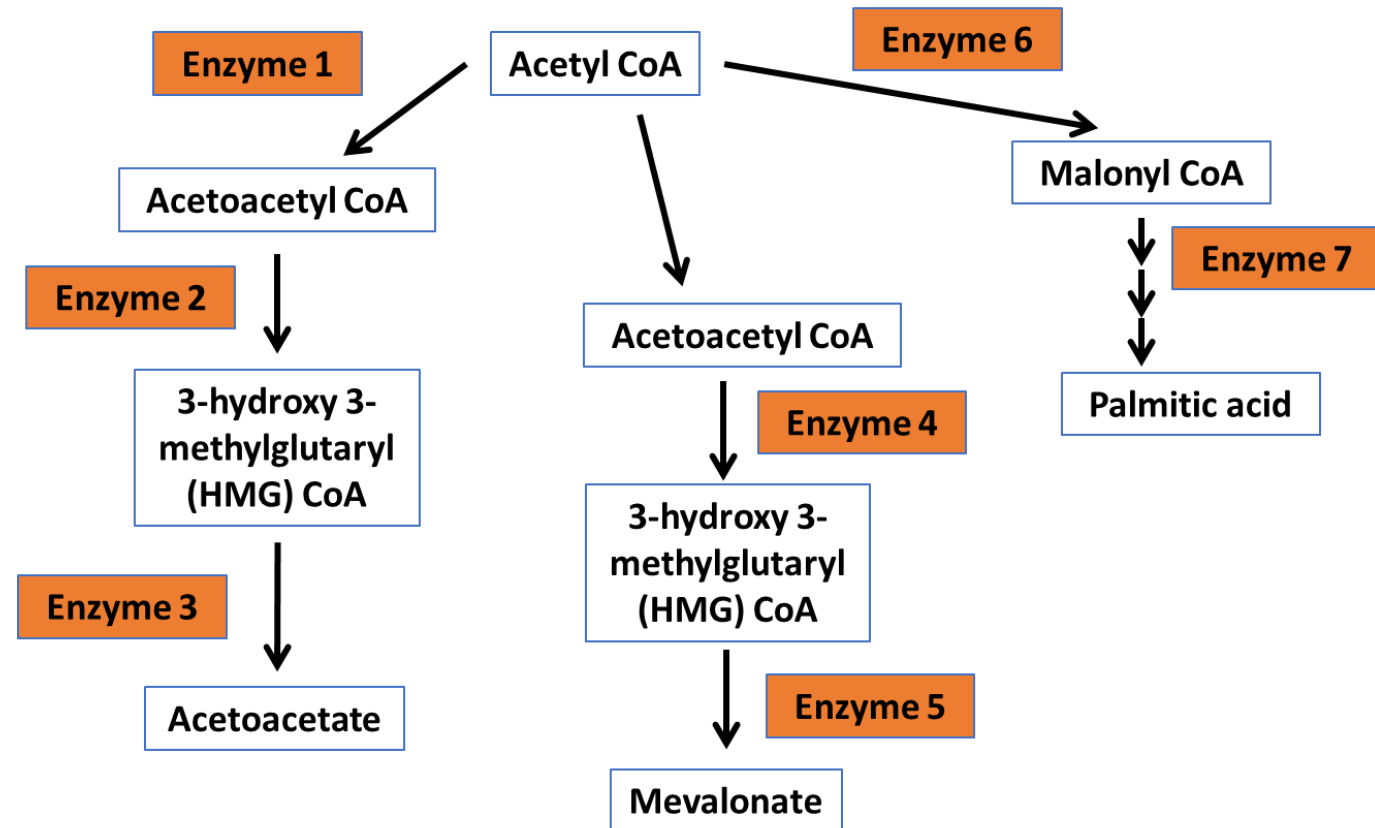
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A 40-year-old man comes to the physician for a routine examination. Laboratory studies show total serum cholesterol level is 270 mg/dL (Ref:<200mg/dL). He is prescribed a drug that competitively inhibits a specific enzyme of intermediary metabolism. Which of the following enzymes is most likely inhibited by this drug?



- A. Enzyme 1
- B. Enzyme 2
- C. Enzyme 3
- D. Enzyme 4
- E. Enzyme 5
- F. Enzyme 6
- G. Enzyme 7

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SOM. MK.I.BPM2.3.DM.2.BCHM.1210; Outline cholesterol synthesis and the role of acetylCoA as a precursor.
SOM. MK.I.BPM2.3.DM.2.BCHM.1215; Describe the role of statins in the management of hypercholesterolemia

A 3-month-old boy is brought to the physician for a follow-up examination. He has a 3-week history of repeated vomiting and intermittent episodes of convulsions. A comprehensive metabolic profile performed on blood and urine samples from the patient. Laboratory studies show severe hyperammonemia and elevated glutamine levels. The levels of citrulline, argininosuccinate, arginine, and blood urea nitrogen are low. Which of the following inherited enzyme deficiencies is most likely in this patient?

- A. Carbamoyl phosphate synthetase-I
- B. Ornithine transcarbamylase
- C. Arginase
- D. Arginosuccinate synthetase
- E. Arginosuccinate lyase

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A 35-year-old man comes to the physician because he has noted gradual increase in difficulty with pain when urinating. Renal stones is suspected, and the patient is provided with a specimen bottle to collect a random mid-stream urine sample for urinalysis. Urinalysis gave negative results across all parameters. Physical examination shows blue black discoloration. When he was disposing of the urine sample sometime after the patient had left, the physician noticed the urine had a dark coloration. Which of the following intermediates is most likely found in the urine of this patient?

- A. Phenylacetate
- B. Homogentisate
- C. Branched chain keto-acids
- D. Methylmalonic acid
- E. Fumarylacetoacetate
- F. Propionate

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A 29-year-old man is brought to the emergency department because of a 2-hour history of vomiting and lethargy. He has a 1-week history of being irregular with his diet and insulin therapy. He has a 4-year history of type 1 diabetes mellitus on treatment with insulin. On examination, he is stuporous, poorly responsive to verbal stimuli and his breath has a sweet, fruity odor. His pulse is 113/min, blood pressure is 95/58 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 93%. Which of the following laboratory findings is most likely in this patient?

- A. Decreased plasma glucose level
- B. Absence of urinary ketone bodies
- C. Increased plasma glucose level
- D. Increased plasma insulin levels
- E. Negative dipstick test for urine glucose
- F. Increased arterial pH
- G. Normal anion gap

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A male newborn delivered at term is found to have heart defects, limb malformation, microcephaly, and ambiguous genitalia. Chromosomal analysis shows a 46,XY karyotype. Targeted gene mutation analysis shows a pathogenic allele resulting in a deficiency of an enzyme involved in the synthesis of cholesterol. Which of the following enzyme-catalyzed reactions is most likely deficient in this patient?

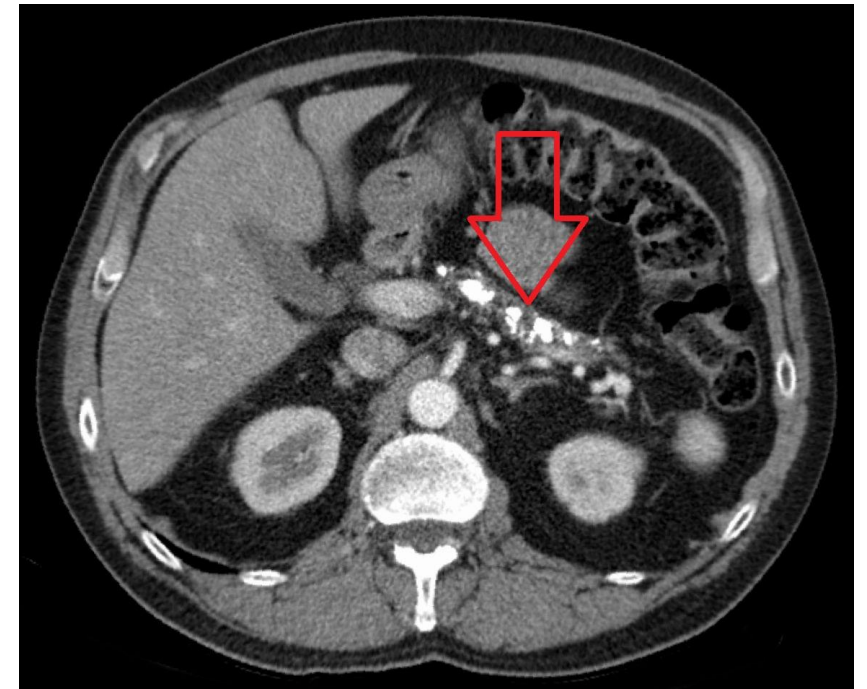
- A. Formation of HMG-CoA
- B. Conversion of HMG-CoA to mevalonate
- C. Formation of isopentenyl pyrophosphate
- D. Formation of farnesyl pyrophosphate
- E. Formation of squalene
- F. Conversion of lanosterol to cholesterol

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A 40-year-old woman comes to the physician because of a 1-year history of intermittent abdominal pain. She says the pain is in the epigastric region and radiates to her back. She consumes alcohol almost daily. Physical examination shows mild tenderness on deep palpation in the epigastric region but without guarding or rebound tenderness. Her pulse is 74/min, respirations are 14/min and temperature is 37.2°C (98.9°F). Laboratory studies show aspartate aminotransferase of 32 U/L (8-40 U/L), alanine aminotransferase of 24 U/L (8-40 U/L), serum amylase of 34 U/L (25-125 U/L), and serum lipase of 37 U/L (10-140 U/L). CT scan of the abdomen is shown. Stool analysis shows increased fat content. Which of the following best explains the stool findings in this patient?

- A. Loss of intestinal mucosal brush border and villi
- B. Decreased synthesis of bile salts
- C. Increased gastric acid secretion
- D. Decreased availability of pancreatic enzymes
- E. Impaired chloride secretion by exocrine glands
- F. Low activity of intestinal disaccharidases

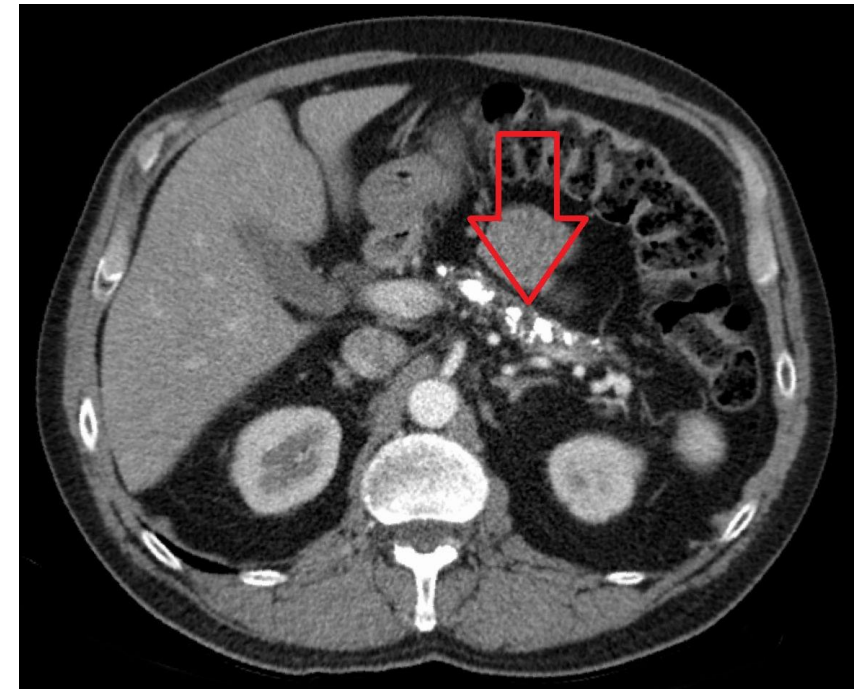


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SOM. MK.I.BPM2.3.DM.3.BCHM.1253; Discuss bile composition and factors facilitating gallstone formation (cholelithiasis) and predict effect of bile duct obstruction on lipid digestion and absorption.

SOM. MK.I.BPM2.3.DM.3.BCHM.1254; Predict consequences of pancreatic disease (cystic fibrosis and pancreatitis) on digestion and discuss basis for dietary management.



A 2-day-old male newborn is brought to the emergency department by his mother because of poor feeding, difficulty breathing and lethargy. His temperature is 36.8°C (98.2°F), blood pressure is 65/40 mmHg, pulse is 143/min, and respirations are 55/min. Physical examination shows generalized hypotonia. Laboratory findings show elevated plasma ammonia, glutamine, and orotic acid; while the blood urea nitrogen (BUN) level is low. Which of the following best describes the mode of inheritance and location of the enzyme deficient in this patient’s condition?

- A. I
- B. II
- C. III
- D. IV
- E. V
- F. VI

	Mode of inheritance	Location of enzyme deficient
I	Autosomal dominant	Mitochondrial
II	Autosomal recessive	Mitochondrial
III	X-linked recessive	Mitochondrial
IV	Autosomal dominant	Cytosolic
V	Autosomal recessive	Cytosolic
VI	X-linked recessive	Cytosolic

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VI	X-linked recessive	Cytosolic

A 5-day-old male newborn is brought to the emergency department by his mother because of a 2-day history of poor feeding, vomiting, and irritability. The mother reports that the diapers smell like burnt sugar. Physical examination shows abnormal posturing and hypertonia. Laboratory evaluation confirms an inborn error of metabolism. Which of the following enzymes most likely requires the same cofactor as the enzyme deficient in this patient?

- A. Acetyl-CoA carboxylase
- B. Amino acid decarboxylase
- C. Phenylalanine hydroxylase
- D. α -ketoglutarate dehydrogenase
- E. Dopamine β -hydroxylase

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SOM. MK.I.BPM2.3.DM.2.BCHM.1234; Differentiate between the inborn errors of amino acid metabolism based on clinical features and laboratory findings. Discuss the significance of vitamin or coenzyme supplementation in inborn errors of amino acid metabolism
SOM. MK.I.BPM2.3.DM.2.BCHM.1235; Outline the catabolic pathway of branched chain amino acids and explain the metabolic basis for Maple syrup urine disease, Methylmalonic aciduria.

A group of scientists observe that fructose consumption increased by almost 50% among US adults in the last 40 years which unlike glucose, is not regulated by insulin. They have observed that high levels of fructose consumption can cause dyslipidemia, impair glucose homeostasis and increase insulin resistance. They review viable drugs to exclusively inhibit fructose transport and no other carbohydrates. Which of the following transporters is most likely being targeted in these drug discovery programmes?

- A. GLUT1
- B. SGLT2
- C. GLUT2
- D. GLUT5
- E. SGLT1

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A 6-day-old male newborn is brought to the emergency department because of repeated episodes of vomiting and lethargy. Physical examination shows tachypnea and hypotonia. Laboratory studies show metabolic acidosis, hyperammonemia and elevated serum methylmalonate. Which of the following nutrients is most likely recommended to be restricted in his diet?

- A. Arginine
- B. Phenylalanine
- C. Tyrosine
- D. Galactose
- E. Fructose
- F. Valine
- G. Phytanic acid
- H. Lactose
- I. Leucine

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SOM. MK.I.BPM2.3.DM.2.BCHM.1220; Distinguish the fates of C- skeletons of amino acids: Glucogenic amino acids, Ketogenic amino acids, Glucogenic & Ketogenic amino acids. Enumerate examples of amino acids in the three groups